Algorithmic Intelligence

Mathematical Optimization from Linear Programming to Quantum Computing

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Applied Algorithmic
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Claude's Cycles

Don Knuth, Stanford Computer Science Department
(28 February 2026; revised 02 March 2026)

Shock! Shock! I learned yesterday that an open problem I'd been working on for several weeks had just been solved by Claude Opus 4.6 — Anthropic's hybrid reasoning model that had been released three weeks earlier! It seems that I'll have to revise my opinions about "generative AI" one of these days. What a joy it is to learn not only that my conjecture has a nice solution but also to celebrate this dramatic advance in automatic deduction and creative problem solving. I'll try to tell the story briefly in this note.

Here's the problem, which came up while I was writing about directed Hamiltonian cycles for a future volume of *The Art of Computer Programming*:

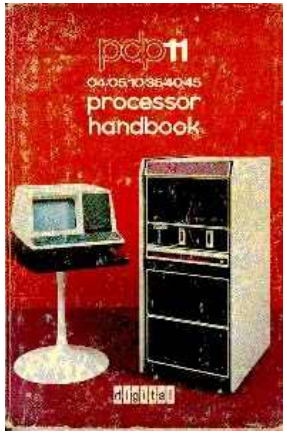
> Consider the digraph with m^3 vertices ijk for $0 \le i, j, k < m$, and three arcs from each vertex, namely to i^+jk, ij^+k, and ijk^+, where $i^+ = (i+1) \bmod m$. Try to find a general decomposition of the arcs into three directed m^3-cycles, for all $m > 2$.

https://cs.stanford.edu/~knuth/papers/claude-cycles.pdf

1970

“Fast algorithms – fast computers”

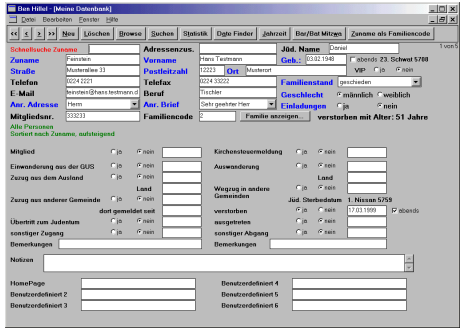
Hardware/Software: What can we compute?



1990

“Once the data is entered, we will get a result”

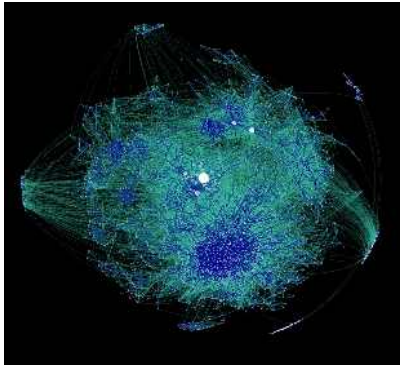
How do we get the data into the computer?

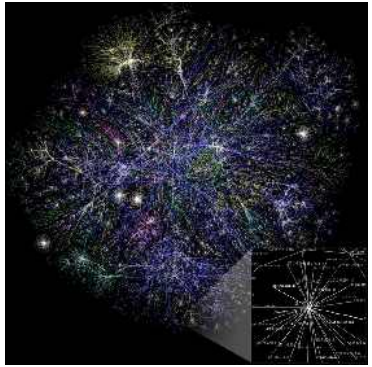


2010

“Computerized information processing is ubiquitous”

How do we find answers to complex questions?





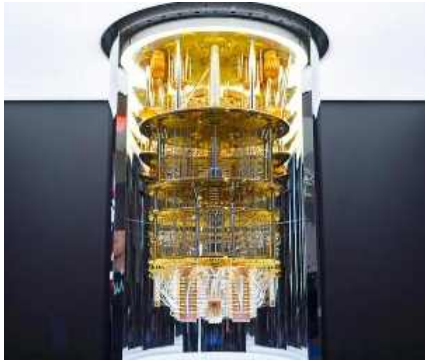
2030

?

?

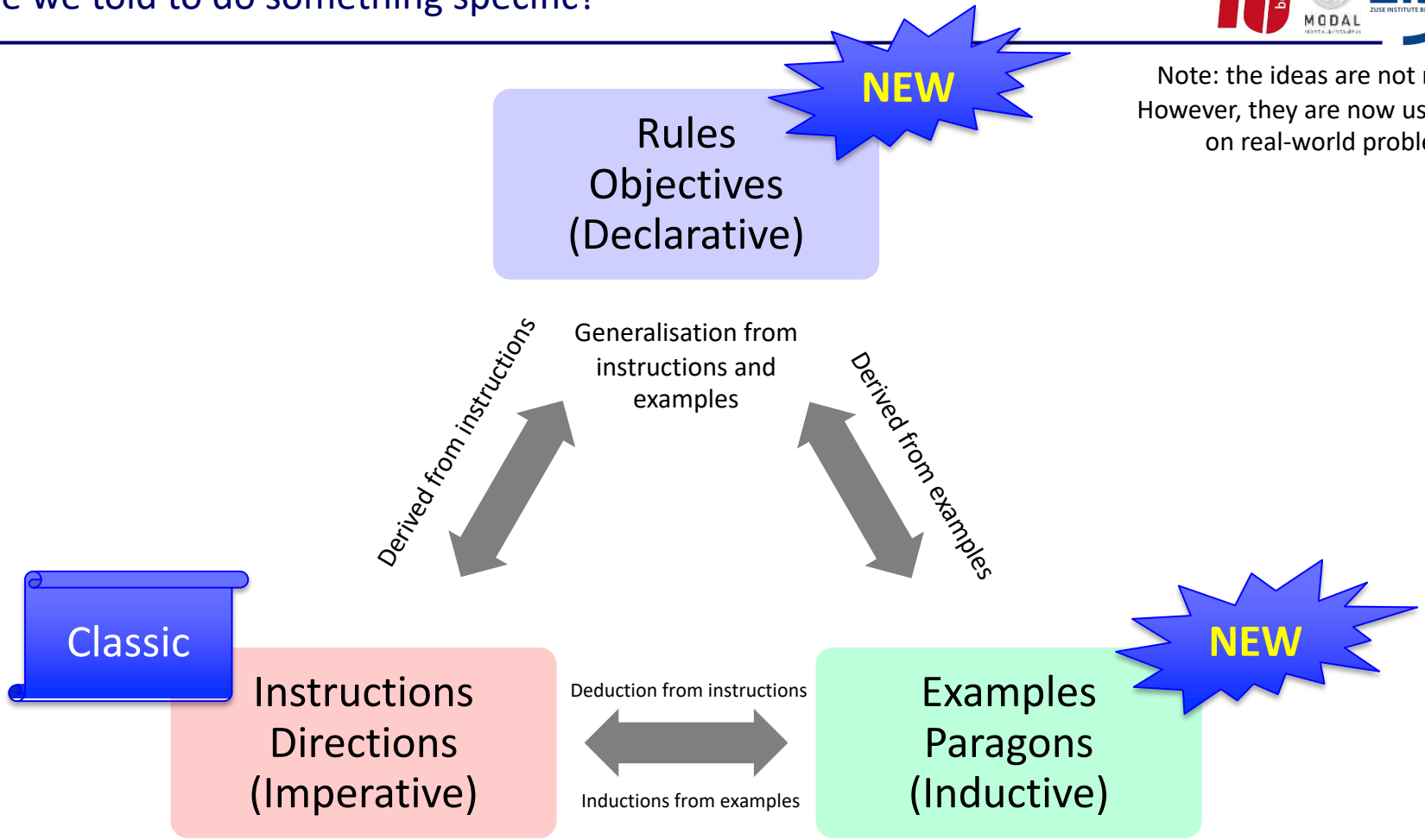
“Quantum enhanced AI systems”

Can we build smart and reliable autonomous systems?



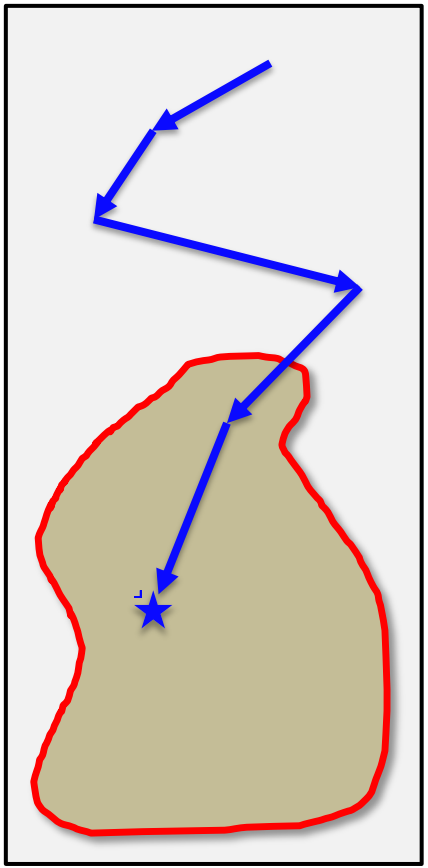
How are we told to do something specific?

Note: the ideas are not new. However, they are now usable on real-world problems.



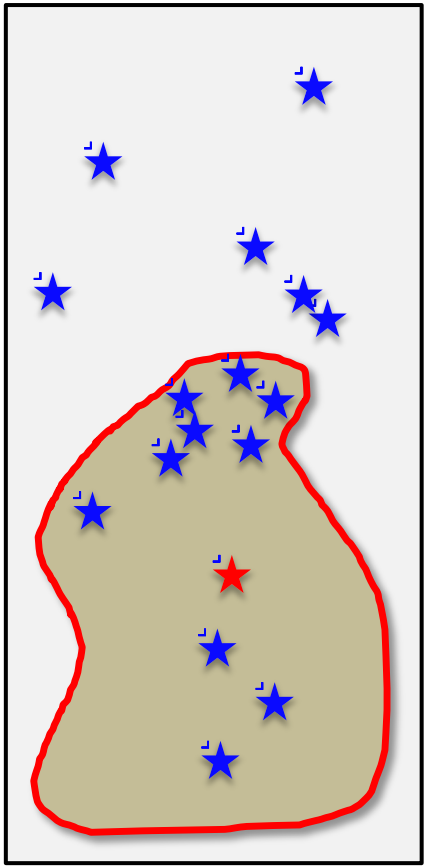
Imperative

How	By detailed instructions
What	Describes a path to the solution
Example	Beat eggs with milk, sugar, salt, flour and mineral water until smooth. Heat a pan with a little oil. Pour a ladle of batter into the pan and swirl briefly to distribute the batter. Fry the pancake on both sides for about 1-2 minutes until brown.
Challenge	Simplest description, fastest implementation
What can go wrong?	Missing steps, missing (special-) cases
Result	Exactly as specified
Explanation	Easy: step by step
Name	Classical programming
	`If x == 5 then` `  print “hello world”`



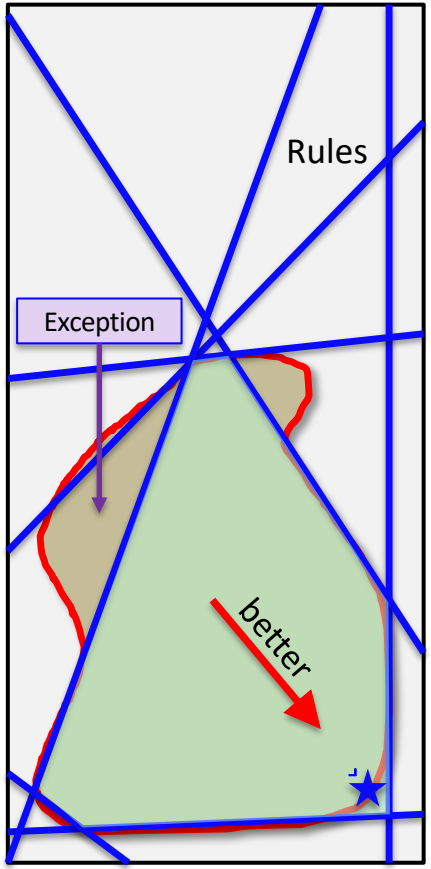
Inductive

How	By example
What	Sampling the solution space
Example	Do it like the others here. Just don't take Kevin as an example, he's a bad role model.
Challenge	Finding enough suitable examples
What can go wrong?	Too few or too bad examples. Borderline cases missing. Poor distribution of examples.
Result	With a certain probability (on average) good.
Explanation	Often difficult
Name	Machine learning Neural Networks
	`model.fit(train_images,` `  train_labels, epochs=10)`



Declarative

How	By rules (laws) and objectives
What	Defining goals and constrains regarding feasible solutions
Example	§Section 263a (1) Whoever, with the intention of obtaining an unlawful pecuniary advantage for himself or a third party, damages the property of another by influencing the result of a data processing operation by incorrect design of the program, by use of incorrect or incomplete data, by unauthorized use of data or otherwise by unauthorized influence on the process, shall be punished by imprisonment for not more than five years or by a fine.
Challenge	Completeness and consistency (of the rules)
What can go wrong?	Missing and contradicting rules
Result	Provable (in-)feasibility and quality
Explanation	Super-human complexity possible
Name	Mathematical Programming
	min $c^T x$ subject to $Ax \leq b, x \in \{0,1\}$



Instruction Paradigms: How to instruct people and computers

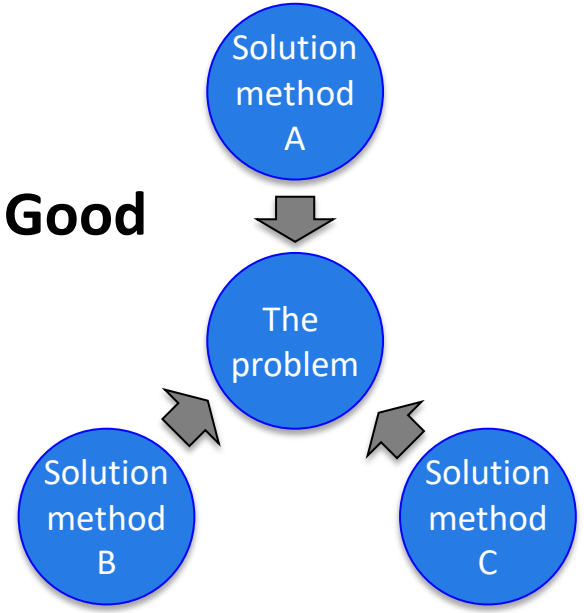
	Imperative	Declarative	Inductive
How	Instructions	Rules, Objectives	Examples
What	Describes a path	Describes a model	Samples the solution space
Name	Classical Programming	Mathematical Programming	Machine Learning / NN
Example	First heat 3 spoons of oil, then pan fry for 10 min, …	Laws, Business rules	Watch me!
	If x == 5 then print “hello world”	min $c^T x$ subject to $Ax \leq b, x \in \{0,1\}$	
Challenge	Fastest implementation	Adequate modelling	Getting enough suitable data
What can go wrong?	Missing cases Missing steps	Missing constraints: unbounded Contradicting constraints: infeasible Wrong objective	Too few or wrong samples (bad distribution, corner cases missing)
If wrong data?	Fails, if not anticipated	Leads easily to infeasibility	Mostly small effect
Code errors?	Likely fails	Often surprisingly small effect	May not converge
Special HW	All classic: CPU, GPU	D-Wave, Fujitsu Digital Annealer	TPU, GPU
Result	Precisely as specified	Provable feasibility (quality)	Probabilistic (average) (good)

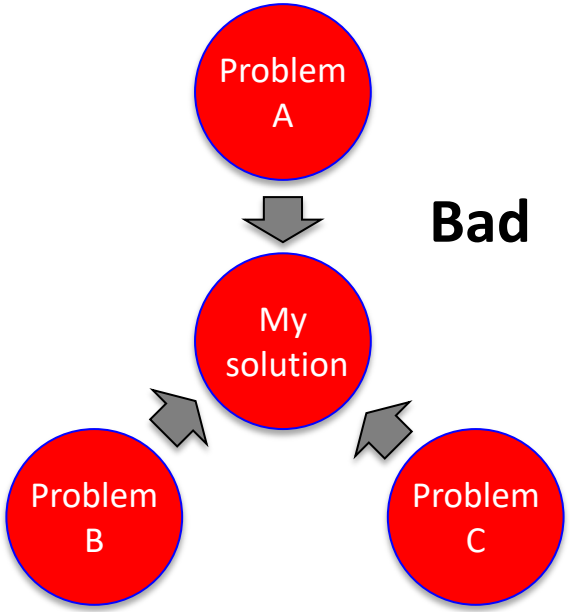
Think and act problem centric

- It is more important to solve the right problem than to solve the problem right.
- Identifying the problem is half of the way to the solution.

"I suppose it is tempting,
if the only tool you have is a hammer,
to treat everything as if it were a nail."

— Abraham Maslow, 1966





My first computer system (1982)

8-bit CPU: Zilog Z80, at 1.76 MHz

Video: Monochrome

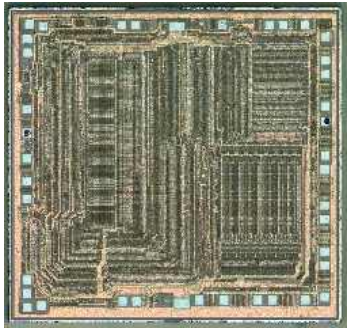
64×16 (uppercase) text, 128×48 block graphics

Composite video output, RF TV signal output

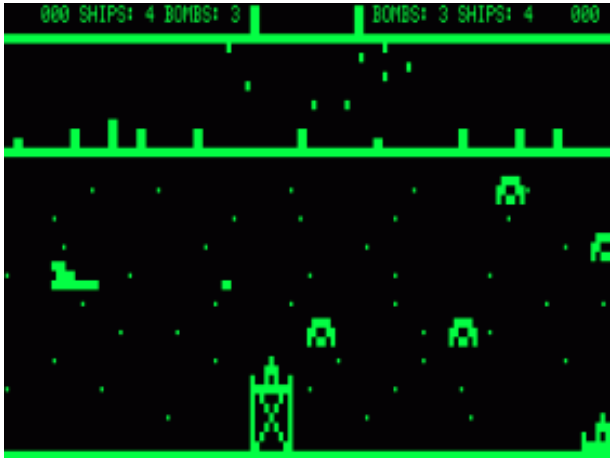
16 KB RAM, expandable to 48 KB

12 KB ROM containing Microsoft LEVEL II BASIC

Storage: Built-in 500 baud cassette deck



Z80 CPU: 8500 transistors



Speed-up of a modern computer regarding neural networks

Here is a comparison between my first computer and a system I currently use.

The calculation is a very rough estimate.

However, it is reasonable to say, that for particular applications, the modern system is about

~400 Billion times faster.

400 Billion times faster means:

80 years become 1 second

	Z80	GH200	speedup
CPU cores	1	72	
L1 cache [KB]		64 + 64	
L2 cache [MB]		72 × 1	
L3 cache [MB]		114	
clockspeed [GHz]	0,00176	3.1	1.761
RAM size [GB]	0,000048	480	10 Mil
FP64 [GFLOPS]	0,0000018	34,000	19 Bil
FP64 TC [GFLOPS]		67,000	
FP32 [GFLOPS]	0,0000061	67,000	11 Bil
FP32 TC [GFLOPS]	0.0000023	989,000*	430 Bil
FP16 TC [GFLOPS]		1,979,000*	
FP8 TC [GFLOPS]		3,958,000*	

*sparse

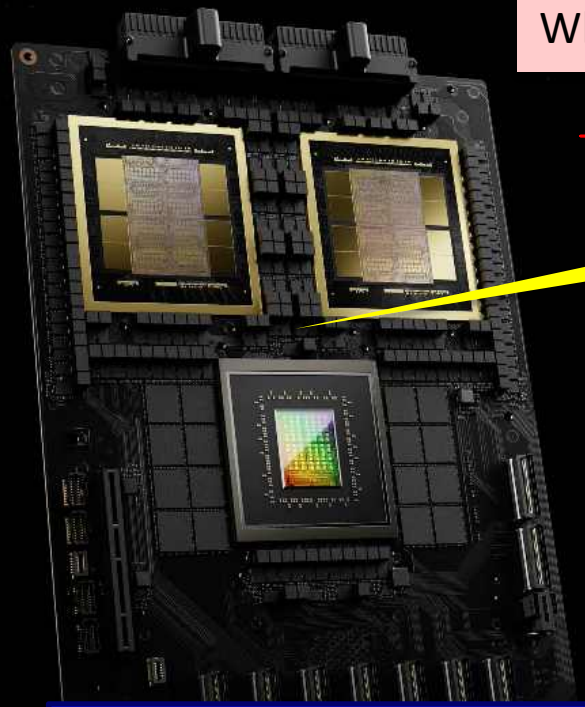
https://www.smxrtos.com/ussw/gofast/gofast_z80_z180_64180.pdf
https://resources.nvidia.com/en-us-grace-cpu/grace-hopper-superchip

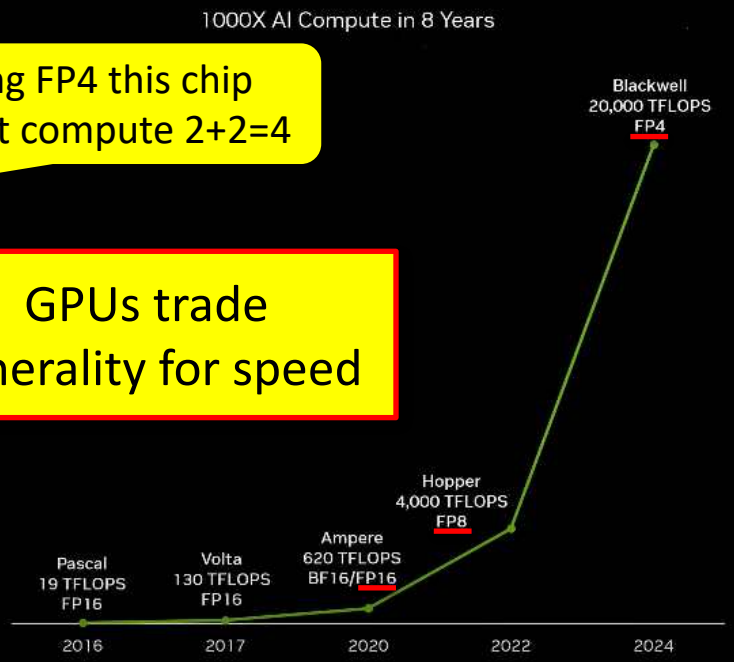
NVIDIA Blackwell: The new $40.000 AI Superchip

Which numbers can be represented exactly with FP4?

Using FP4 this chip cannot compute 2+2=4

GPUs trade generality for speed





However, 1.58 bits are enough for inference. doi 10.5220/0013382400003890

FP4

```
-Inf
-3.0
-2.0
-1.5
-1.0
-0.5
 0.0
 0.5
 1.0
 1.5
 2.0
 3.0
 Inf
 NaN
```

That's all folks!

Not in our direction: NVIDIA GPU Performance in TeraFLOPS

Architecture GPU-Name	Blackwell Ultra B300	Blackwell B200	Hopper H200	Hopper H100	Ampere A100
FP64	**1.2**	**37**	**34**	**34**	9.7
FP64 Tensor Core	**1.2**	**37**	**67**	**67**	19.5
FP32	75	75	67	67	19.5
FP32 Tensor Core	2,200	2,200	989	989	312
FP16/BF16 Tensor Core	4,500	4,500	2,000	2,000	624
FP8/FP6 Tensor Core	9,000	9,000	4,000	4,000	-
FP4 Tensor Core (Dense)	14,000	9,000	-	-	-
FP4 Tensor Core (Sparse)	18,000	18,000	-	-	-

Tensor operation $C = A \cdot B + C$ (GEMM 8x4)

https://www.exxactcorp.com/blog/hpc/comparing-nvidia-tensor-core-gpus

Nemotron-3-super with opencode or openclaw

opencode

Knapsack Rust: bug fixes and performance review in 11-knapsack/rust/...

4. Memory Optimization (Main Request):
   - Original: O(n*C) space for full DP table (n+1) × (C+1)
   - Optimized: O(C) space for DP computation using only two rows (previous/current)
   - Trade-off: Still uses O(n*C) boolean tracking matrix for solution reconstruction
   - Result: ~73% memory reduction for typical cases (tested with n=100, C=1000: 404KB → 109KB)

Verification:

- Tested with original example: produces identical correct output (items 1,3; total value 12)
- Tested with large random input (n=100, C=1000): same optimal solution and performance
- Verified all error handling works correctly with various invalid inputs
- Both implementations maintain O(n*C) time complexity

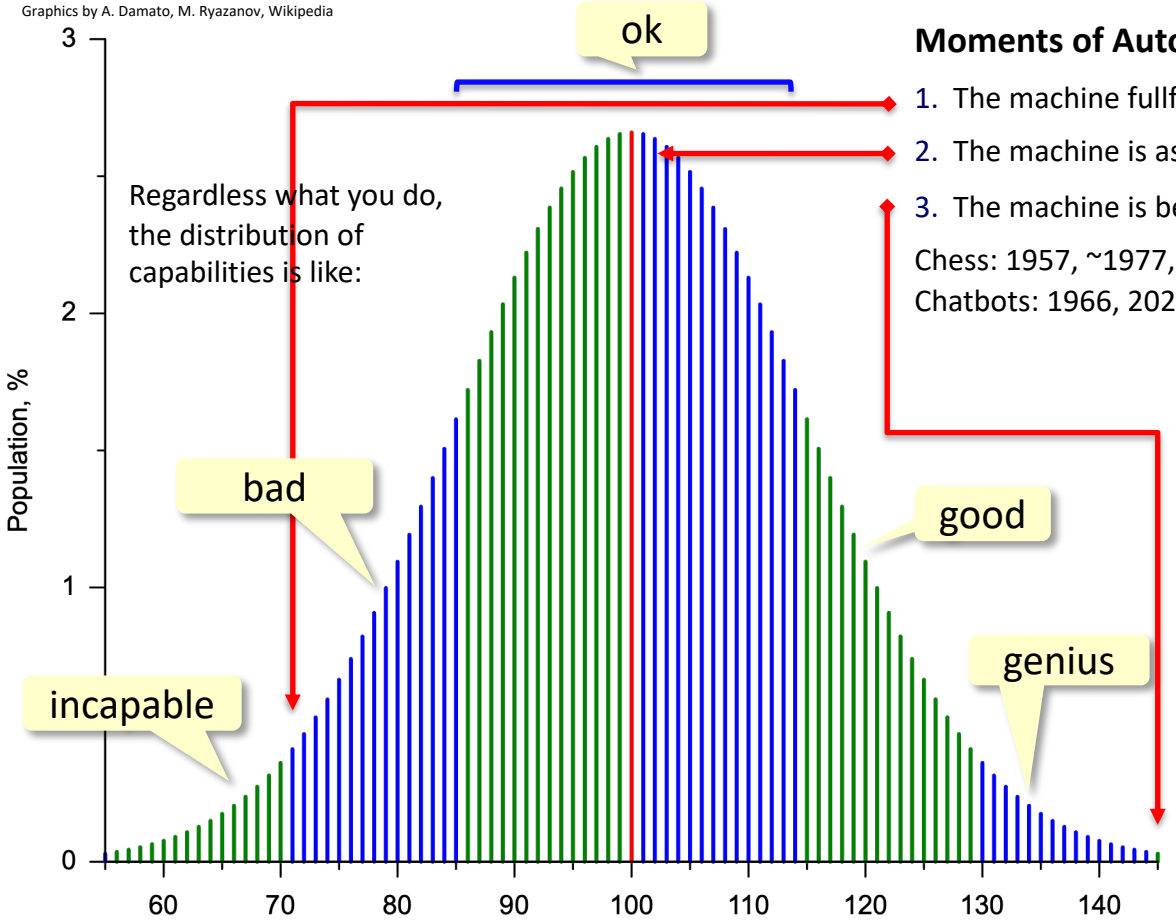
The optimized implementation now uses significantly less memory for the DP computation while maintaining the same correctness and improving robustness through better error handling.

Build · Nemotron 3 Super Free · 10m 25s

Geändert 5 Dateien

How good a machine needs to be to replace a human?

Graphics by A. Damato, M. Ryazanov, Wikipedia



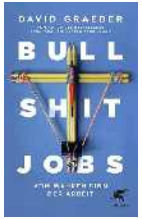
Moments of Automatization

1. The machine fullfills the task first time formally correct.
2. The machine is as good or better than the average person.
3. The machine is better than the best human.

Chess: 1957, ~1977, 1997, Go: 1968, ~2009, 2017,
Chatbots: 1966, 2021, ...

From 2, the „Good Performers“ working computer aided can replace everone left from them.

Fully automated comes much later, shortly before 3..



Before you start to worry, read first this book 😉

Computational Irreducibility

▷ S. Wolfram: And in the end there's just a fundamental tension between learnability and computational irreducibility. Learning involves in effect compressing data by leveraging regularities. But computational irreducibility implies that ultimately there's a limit to what regularities there may be.

https://en.wikipedia.org/wiki/Busy_beaver

Collatz Conjecture:

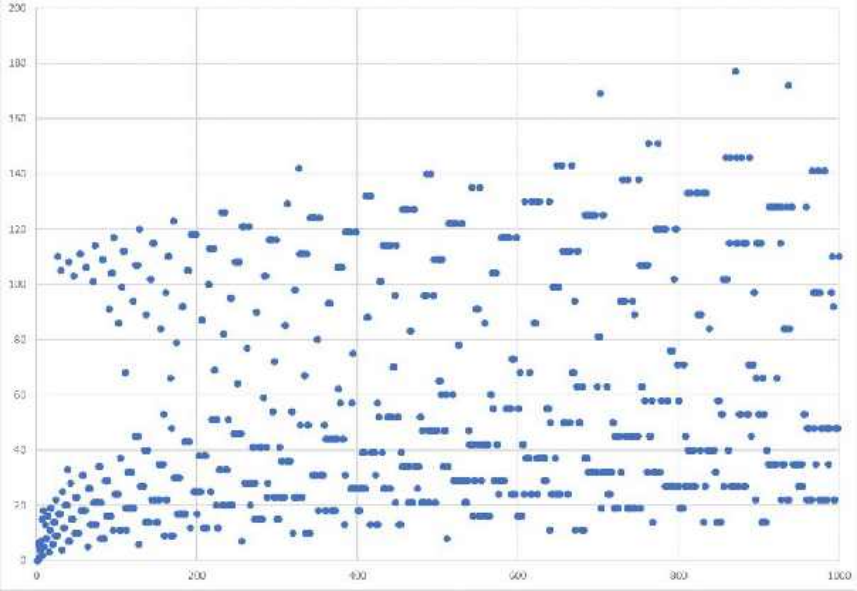
Given $n \in \mathbb{N}_+$

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n+1 & \text{otherwise} \end{cases}$$

$$a_i = \begin{cases} n & \text{for } i = 0, \\ f(a_{i-1}) & \text{for } i > 0 \end{cases}$$

This process will reach the number 1, for any n, i.e., for each n, there is some i with $a_i = 1$.

https://de.wikipedia.org/wiki/Collatz-Problem



By Onecoldruben - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=91983030

Algorithms + Data Structures = Programs

—Niklaus Wirth, 1976

Instruction Paradigms ✕ Computing = Algorithmic Intelligence

The Bitter Lesson, Rich Sutton, March 13, 2019

This is a big lesson. As a field, we still have not thoroughly learned it, as we are continuing to make the same kind of mistakes. To see this, and to effectively resist it, we have to understand the appeal of these mistakes. We have to learn the bitter lesson that building in how we think we think does not work in the long run. The bitter lesson is based on the historical observations that 1) AI researchers have often tried to build knowledge into their agents, 2) this always helps in the short term, and is personally satisfying to the researcher, but 3) in the long run it plateaus and even inhibits further progress, and 4) breakthrough progress eventually arrives by an opposing approach based on scaling computation by search and learning. The eventual success is tinged with bitterness, and often incompletely digested, because it is success over a favored, human-centric approach.

One thing that should be learned from the bitter lesson is the great power of general purpose methods, of methods that continue to scale with increased computation even as the available computation becomes very great. The two methods that seem to scale arbitrarily in this way are *search* and *learning*. http://www.incompleteideas.net/IncIdeas/BitterLesson.html

Top500 Nov 2025 http://top500.org

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray Shasta, AMD [illegible] HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, Rib Supernode, GC 3GHz, NVIDIA GH200 Superchip, Quad Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794
5	Azure - Microsoft ND[illegible] Platform, [illegible] NVIDIA H100, NVIDIA InfiniBand NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
6	HPC6 - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.8, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461

The Top 10 computers in the world all have at least 1 Million computing “cores” (CPU+GPU).

A good question to ask, if you visit one of them is:

How many (non benchmark) applications do you have, which use the whole system?

There are still programs which run too long, there are problems that can’t be solved. How can we utilize the available computing capability to solve whatever our problem is? Or: Why is this soooo difficult?

https://en.wikipedia.org/wiki/TOP500

https://blogs.nvidia.com/blog/2021/11/09/cuquantum-world-record/

NVIDIA Sets World Record for Quantum Computing Simulation With cuQuantum

Theoretical computer scientist mean something different by "efficiently"
+ by this definition just finding a solution is **not** "solving" it.

Driving toward that future, NVIDIA created the largest ever simulation solving the MaxCut problem using cuQuantum, our SDK for accelerating GPU.

Not the optimal solution.
We need **0.2 s** on a workstation to find this quality solution.
2 days to find the optimal solution,
3 more days to prove optimality.

In the math world, MaxCut is often cited as an example of an optimization problem no known computer can solve efficiently. MaxCut algorithms are used to design large computer networks, find the optimal layout of chips with billions of silicon pathways and explore the field of statistical physics.

[...]

We used the cuTensorNet library in cuQuantum running on NVIDIA's in-house supercomputer, Selene, to simulate a quantum algorithm to solve the MaxCut problem. Using 896 GPUs to simulate 1,688 qubits, we were able to solve a graph with a whopping 3,375 vertices. That's 8x more qubits than the previous largest quantum simulation.

Our solution was also highly accurate, reaching 96% of the best-known answer. We set this new record with an algorithm developed by NVIDIA researchers and an open-source framework.

QuBowl:
An exact solver for QUBO and Max-Cut

Faster exact solution of sparse MaxCut and QUBO problems

Daniel Rehfeldt, Thorsten Koch, Yuji Shinano

doi: 10.1007/s12532-023-00236-6

Mathematical Programming Computation
https://doi.org/10.1007/s12532-023-00236-6

FULL LENGTH PAPER

Faster exact solution of sparse MaxCut and QUBO problems

Daniel Rehfeldt[1] · Thorsten Koch[1,2] · Yuji Shinano[1]

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Abstract
The maximum-cut problem is one of the fundamental problems in combinatorial optimization. With the advent of quantum computers, both the maximum-cut and the equivalent quadratic unconstrained binary optimization problem have experienced much interest in recent years. This article aims to advance the state of the art in the exact solution of both problems—by using mathematical programming techniques. The main focus lies on sparse problem instances, although also dense ones can be solved. We enhance several algorithmic components such as reduction techniques and cutting-plane separation algorithms, and combine them in an exact branch-and-cut solver. Furthermore, we provide a parallel implementation. The new solver is shown to significantly outperform existing state-of-the-art software for sparse maximum-cut and quadratic unconstrained binary optimization instances. Furthermore, we improve the best known bounds for several instances from the 7th DIMACS Challenge and the QPLIB, and solve some of them (for the first time) to optimality.

Keywords Maximum-cut · Quadratic unconstrained binary optimization · Branch-and-cut

Mathematics Subject Classification 90C27 · 90C10

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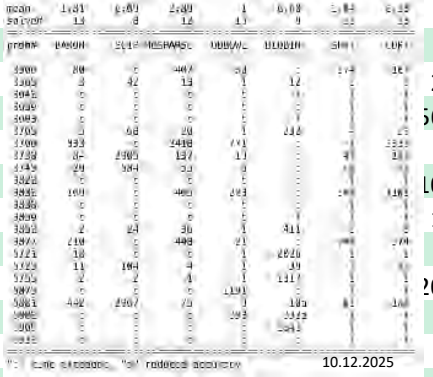
Springer

Hans Mittelmann: Nonconvex QUBO-QPLIB Benchmark, 10.07.2023

prob#	Baron	SCIP	McSparse	Octeract	Gurobi	QuBowl	BiqBin	SHOT	ABS2
3506	87		407	89	165	102		789	0.047
3565	2	52	13	2	5	1	12	3	0.028
3642							f		0.063
3650									0.219
3693							f		*0.899*
3705	5	94	20	7	6	1	232	6	0.043
3706	881		2418	899	2357	984			0.096
3738	58	2753	137	73	94	29		197	0.096
3745	24	442	55	29	14	12		24	0.043
3822									0.051
3832	218		465	244	260	245			0.252
3838									7.971
3850							f		1.844
3852					2	2	411	2	0.020
3877					322	36			0.044
5721					1	1	2026	1	*7.509*
5725					7	1	39	1	0.122
5755					1	1	1317	1	1.061
5875						1720	1150		0.002
5881					47	29	10	64	0.001
5882						607	232		0.000
5909							1842		0.019
5922									0.022
mean	1.85	5.65	2.44	1.80	1.46	1.00	5.31	2.06	
solved	12	8	12	12	13	15	10	11	



QPLIB A Library of Quadratic Programming Instances

https://qplib.zib.de

http://plato.asu.edu/ftp/qubo.html

The codes were run on an AMD Ryzen 9 5900X (12 cores, 128GB) on the 23 unconstrained binary problems from QPLIB. All problems were solved GLOBALLY. Times given are elapsed times in seconds, time limit 1hr; 12 threads. Shifted and scaled geometric mean of runtimes:.

Baron-23.6.22	https://www.minlp.com/baron
SCIP-8.0/cplex	http://scipopt.org
OCTERACT-4.7.1/cplex	https://octeract.com
Gurobi-10.0.1	http://gurobi.com
QuBowl	https://arxiv.org/pdf/2202.02305.pdf
Biqcrunch-2	https://biqcrunch.lipn.univ-paris13.fr
BiqBin	http://www.biqbin.eu
McSparse-2.0	http://mcsparse.uni-bonn.de
SHOT-1.1/gurobi	https://shotsolver.dev/shot
ABS/Amplify	https://amplify.fixstars.com/en

ABS2 on RTX4090 (red = suboptimal solution, gap 0.17, 0.09)

Max-Cut

Graph formulation $G = (V, E, w)$

$$\max_{S,T} \sum_{i \in S, j \in T} w_{ij} \quad \text{with } S \subset V, T \subset V, S \cap T = \emptyset, S \cup T = V$$

Ising formulation

$$\max \frac{1}{4} \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(1 - x_i x_j), x_k \in \{1, -1\} \text{ for } k \in \{1, \dots, n\}$$

Binary Linear Programing formulation:

$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} z_{ij}$$
$$z_{ij} \leq x_i + x_j$$
$$z_{ij} \leq 2 - (x_i + x_j)$$
$$x_k \in \{0,1\}$$
$$z_{ij} \in \{0,1\}$$

Binary Quadratic Programing formulation:

$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(x_i + x_j - 2x_i x_j)$$
$$x_i \in \{0,1\}$$

Can be written as:

$$\min_{x \in \{0,1\}^n} x^T Q x$$

QUBO

QUBO : Quadratic Unconstraint Binary Optimization
UBQP : Unconstrained Binary Quadratic Program
(BIQ : Binary Integer Quadratic problem)

$$\min_{x \in \{0,1\}^n} x^T Q x$$

- x is a vector of binary variables, Q is a square $n \times n$ matrix of constants
- Since QUBOs are unconstraint, any 0/1 vector is a feasible solution
- All QUBOs can be brought to the form where Q is symmetric or upper triangular
- Solving QUBO (in general) is NP-hard
- Since x is binary, $x_i = x_i^2$ holds $\Longrightarrow$ The coefficients of the linear terms of the objective function correspond to the diagonal entries of Q

BIP

$$\min_{x \in \{0,1\}^n} c^T x$$
$$\text{s.t. } Ax \leq b$$

QUBO

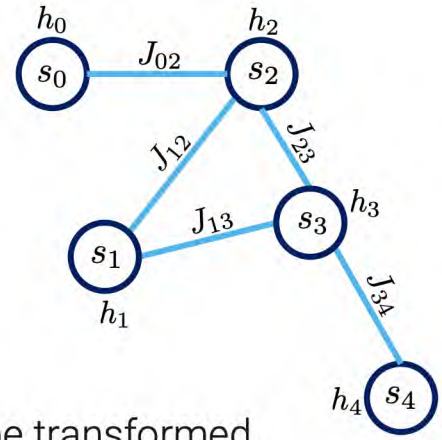
$$\min_{x \in \{0,1\}^n} c^T x^2 + P(Ax + Is - b)^2$$

BIPs can be reformulated as QUBOs by putting the constraints into the objective with a penalty term P. The penalty should be zero if and only if the constraint is fulfilled.

Glover, Kochenberger, Du (2019):
A Tutorial on Formulating and Using QUBO Models
arXiv:1811.11538

Ising Hamiltonian

- Quantum annealing is a special-purpose device that finds the minimum energy of an Ising Hamiltonian heuristically



Linear terms (biases, local fields) | Quadratic terms (couplers) | Spins

$$H_{\text{Ising}} = \sum_{i \in V} h_i s_i + \sum_{\{i,j\} \in E} J_{ij} s_i s_j, \quad s_i \in \{-1, +1\}$$

- A Quadratic Unconstrained Binary Optimization (QUBO) problem can be transformed into an Ising Hamiltonian with a simple algebraic manipulation:

$$x^T Q x \xrightarrow[s_i \in \{-1,1\}]{x_i = \frac{s_i + 1}{2}} H_{\text{Ising}}$$

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Some more notes on QUBOs

- In theory possible to model all ILP and SAT, but not MILP
- **Number of constraints is not important, only the number of Variables!**
- However, a high number of constraints will increase the magnitude of the coefficients.
- Constraint with large support result in dense Q, cardinality constraint is worst
- While most available software works on dense instances this limits the problem size dramatically
- General $P(Ax-b)^T(Ax-b)$. Beware numerical trouble!
- QUBO is unconstraint and pure binary, nearly all heuristic ideas work nicely
- Preprocessing is limited compared to ILP/SAT

Clause #	Clause	Quadratic penalty
1	$x_1 \vee x_2$	$1 - x_1 - x_2 + x_1x_2$
2	$x_1 \vee \bar{x}_2$	$x_2 - x_1x_2$
3	$\bar{x}_1 \vee x_2$	$x_1 - x_1x_2$
4	$\bar{x}_1 \vee \bar{x}_2$	x_1x_2
5	$\bar{x}_1 \vee x_3$	$x_1 - x_1x_3$
6	$\bar{x}_1 \vee \bar{x}_3$	x_1x_3
7	$x_2 \vee \bar{x}_3$	$x_3 - x_2x_3$
8	$x_2 \vee x_4$	$1 - x_2 - x_4 + x_2x_4$
9	$\bar{x}_2 \vee x_3$	$x_3 - x_2x_3$
10	$\bar{x}_2 \vee \bar{x}_3$	x_2x_3
11	$x_3 \vee x_4$	$1 - x_3 - x_4 + x_3x_4$
12	$\bar{x}_3 \vee \bar{x}_4$	x_3x_4

Glover, Kochenberger, Du (2019):
A Tutorial on Formulating and Using QUBO Models
arXiv:1811.11538

- Impossible to distinguish between feasibility and optimization. When to stop?
- Best for problems where there is some “natural” quadratic formulation. (however, e.g., QAP is dense)

Quantum Optimization: Potential, Challenges, and the Path Forward

nature reviews physics https://doi.org/10.1038/s42254-024-00770-9

Review article Check for updates

Challenges and opportunities in quantum optimization

Amira Abbas[1,2], Andris Ambainis[3], Brandon Augustino[4], Andreas Bärtschi[5], Harry Buhrman[1,2,6], Carleton Coffrin[7], Giorgio Cortiana[8], Vedran Dunjko[9], Daniel J. Egger[10], Bruce G. Elmegreen[11], Nicola Franco[12], Filippo Fratini[13], Bryce Fuller[14], Julien Gacon[10,15], Constantin Gonciulea[16], Sander Gribling[17], Swati Gupta[4], Stuart Hadfield[18,19], Raoul Heese[20], Gerhard Kircher[13], Thomas Kleinert[21], Thorsten Koch[22,23], Georgios Korpas[24,25,26], Steve Lenk[27,28], Jakub Marecek[28], Vanio Markov[16], Guglielmo Mazzola[29], Stefano Mensa[30], Naeimeh Mohseni[8], Giacomo Nannicini[31], Corey O'Meara[8], Elena Peña Tapia[10], Sebastian Pokutta[32,33], Manuel Proissl[10], Patrick Rebentrost[34], Emre Sahin[30], Benjamin C. B. Symons[30], Sabine Tornow[35], Víctor Valls[36], Stefan Woerner[10], Mira L. Wolf-Bauwens[10], Jon Yard[37], Sheir Yarkoni[38], Dirk Zechiel[21], Sergiy Zhuk[36] & Christa Zoufal[10]

Quantum computers have demonstrable ability to solve problems at a scale beyond brute-force classical simulation.

Interest in quantum algorithms has developed in many areas, particularly in relation to mathematical optimization — a broad field with links to computer science and physics.

In this Review, we aim to give an overview of quantum optimization. Provably exact, provably approximate and heuristic settings are first explained using computational complexity theory, and we highlight where quantum advantage is possible in each context.

Then, we outline the core building blocks for quantum optimization algorithms, define prominent problem classes and identify key open questions that should be addressed to advance the field. We underscore the importance of benchmarking by proposing clear metrics alongside suitable optimization problems, for appropriate comparisons with classical optimization techniques, and discuss next steps to accelerate progress towards quantum advantage in optimization.

Quantum Supremacy and Quantum Utility / Advantage

https://www.quantamagazine.org/john-preskill-explains-quantum-supremacy-20191002

In 2012, I proposed the term "**quantum supremacy**" to describe the point where quantum computers can do things that classical computers can't, regardless of whether those tasks are useful.

https://scottaaronson.blog/?p=4317

Often abbreviated to just "quantum supremacy," the term refers to the use of a quantum computer to solve ***some*** well-defined set of problems that would take orders of magnitude longer to solve with any currently known algorithms running on existing classical computers—and not for incidental reasons, but for reasons of asymptotic quantum complexity. The emphasis here is on being as sure as possible that the problem *really was* solved quantumly and *really is* classically intractable, and ideally achieving the speedup *soon* (with the noisy, non-universal QCs of the present or very near future). If the problem is also *useful* for something, then so much the better, but that's not at all necessary. The Wright Flyer and the Fermi pile weren't useful in themselves.

https://www.quantumcomputinginc.com/blog/quantum-advantage

Quantum utility (advantage) refers to the demonstrated and measured success to process a real-world problem faster on a quantum computer than on a classic computer. […]

Utility / Advantage means you took a problem from a real-world scenario and ran it on a quantum computer with faster, better results. Real-world, as in real data, real problem, real data volumes with regard to variables, constraints and goals.

Note: "real" probably includes solving the *n*-Queens problem on $17 \times 17 \times 17$ board.

Supremacy: **Huge gain on anything**

Utility : **Any gain on something**

John Martinis: **Important Qubit Metrics**

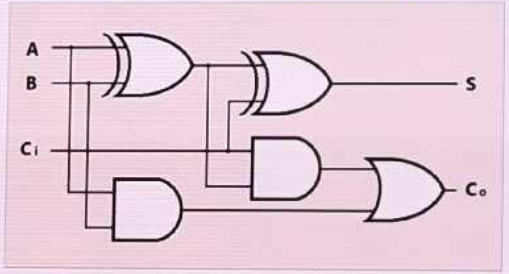
1. Qubit errors (2Q)
2. Number of qubits, scalability
   a) Do errors increase with number?
3. Connectivity: more is better
   a) SC: 4 (Google), 2.5 (IBM)
   b) Ion: All-to-all, for less than ~30 qubits
   (Classical logic does not have all-to-all)
4. Speed
   a) Factor 1000 or more
   b) 1000x faster is 1000x less expensive (all equal)
   c) Fundamental need for re-calibration

Slide from John Martinis

Effect of Qubit Errors (for target)

Classical (CMOS)

space →

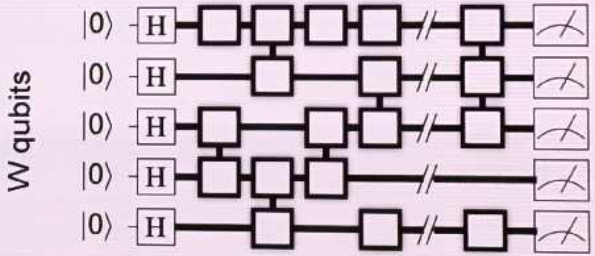


$N_{logic} \sim 1/err_{logic}$

1960's: $err = 10^{-3}$ TTL
2020: $err = 10^{-11}$ M1

Quantum

time (depth D) →



$W\,D \sim 1/err_{gate}$ = coherence time / gate time

W = 100 $err = 10^{-2}$ D ~ 1
$err = 10^{-4}$ D ~ 100

Nvidia Analyst day 2024

„If you said 15 years for very useful quantum computers, that would probably be on the early side.
If you said 30, it's probably on the late side. But if you picked 20, I think a whole bunch of us would believe it.“

– Jensen Huang, CEO Nvidia, Analyst day 2024

https://www.cnbc.com/2025/01/08/quantum-stocks-like-rigetti-plunge-after-nvidias-huang-says-the-computers-are-15-to-30-years-away.html

Von 總統府 - https://www.flickr.com/photos/presidentialoffice/53319959776/, CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=148902092

Quantum Optimization Benchmark Library -- The Intractable Decathlon

Thorsten Koch, David E. Bernal Neira, Ying Chen, Giorgio Cortiana, Daniel J. Egger, Raoul Heese, Narendra N. Hegade, Alejandro Gomez Cadavid, Rhea Huang, Toshinari Itoko, Thomas Kleinert, Pedro Maciel Xavier, Naeimeh Mohseni, Jhon A. Montanez-Barrera, Koji Nakano, Giacomo Nannicini, Corey O'Meara, Justin Pauckert, Manuel Proissl, Anurag Ramesh, Maximilian Schicker, Noriaki Shimada, Mitsuharu Takeori, Victor Valls, David Van Bulck, Stefan Woerner, Christa Zoufal

Through recent progress in hardware development, quantum computers have advanced to the point where benchmarking of (heuristic) quantum algorithms at scale is within reach. Particularly in combinatorial optimization -- where most algorithms are heuristics -- it is key to empirically analyze their performance on hardware and track progress towards quantum advantage. To this extent, we present ten optimization problem classes that are difficult for existing classical algorithms and can (mostly) be linked to practically-relevant applications, with the goal to enable systematic, fair, and comparable benchmarks for quantum optimization methods. Further, we introduce the Quantum Optimization Benchmark Library (QOBLIB) where the problem instances and solution track records can be found. The individual properties of the problem classes vary in terms of objective and variable type, coefficient ranges, and density. Crucially, they all become challenging for established classical methods already at system sizes ranging from less than 100 to, at most, an order of 100,000 decision variables, allowing to approach them with today's quantum computers. We reference the results from state-of-the-art solvers for instances from all problem classes and demonstrate exemplary baseline results obtained with quantum solvers for selected problems. The baseline results illustrate a standardized form to present benchmarking solutions, which has been designed to ensure comparability of the used methods, reproducibility of the respective results, and trackability of algorithmic and hardware improvements over time. We encourage the optimization community to explore the performance of available classical or quantum algorithms and hardware platforms with the benchmarking problem instances presented in this work toward demonstrating quantum advantage in optimization.

Comments: 64 pages, 20 figures. Link to QOBLIB repository: this https URL
Subjects: **Quantum Physics (quant-ph)**; Combinatorics (math.CO)
Cite as: arXiv:2504.03832 **[quant-ph]**
(or arXiv:2504.03832v1 **[quant-ph]** for this version)
https://doi.org/10.48550/arXiv.2504.03832

https://github.com/ZIB-AOPT/QOBLIB

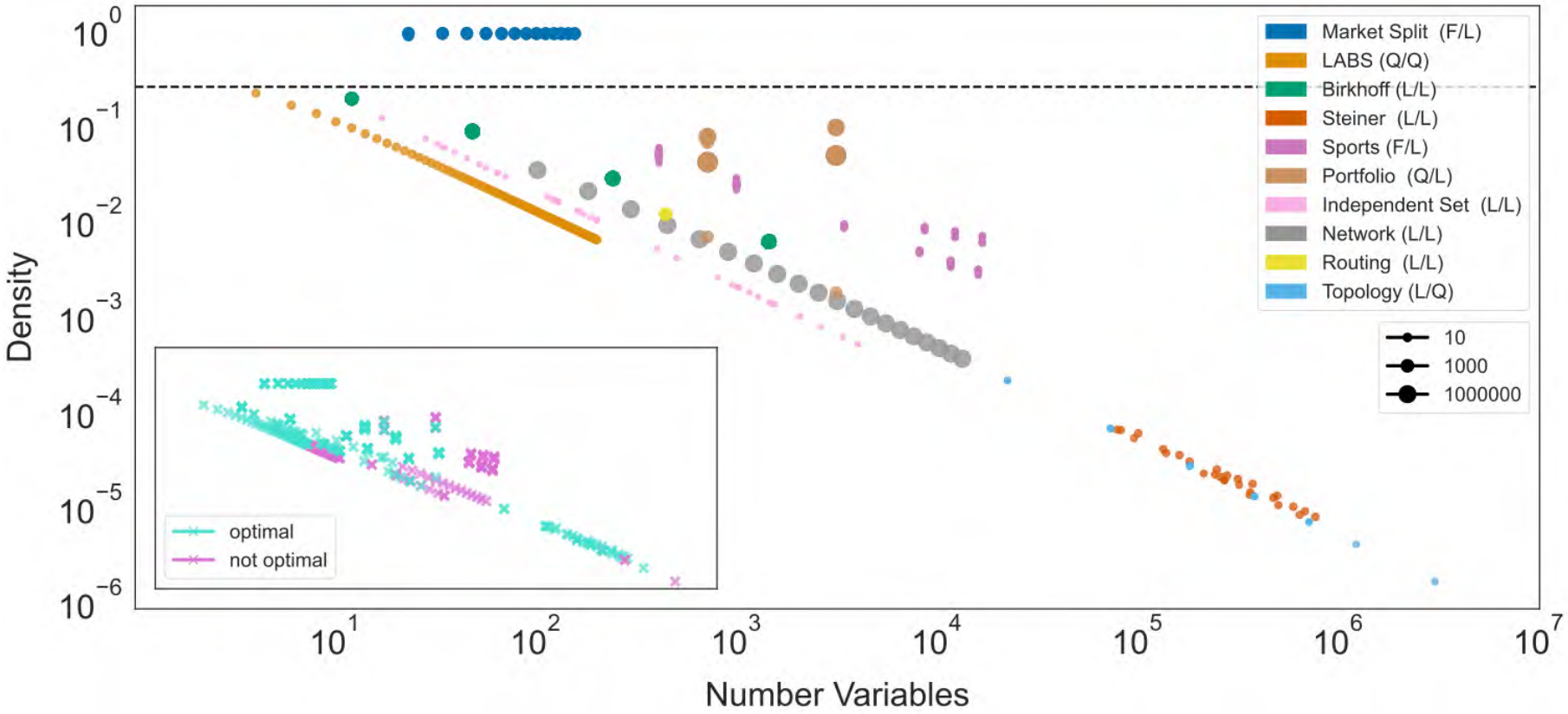
Instance properties

	MIP							QUBO		
Problem	**Type**	**Dense**	**#Vars**	**Coeffs**	**Constr**	**Feas**	**Bin**	**Dense**	**#Vars**	**Coeffs**
Market Split	F / L	d	78	48	1	no	yes	d	70	$\sim 5 \cdot 10^6$
LABS	Q / Q	s	81	4	1	yes	yes	s	820	$\sim 4 \cdot 10^4$
Birkhoff	L / L	s	240	10^4	3	yes	no	d	3,480	$\sim 3 \cdot 10^{10}$
Stein	L / L	s	423,360	3	9	no	yes	-	-	-
Sports	F / L	s	8,608	2	> 10	no	no	s	11, 791	$\sim 4.5 \cdot 10^3$
Portfolio	Q / L	s*	690	$\sim 3 \cdot 10^4$	2	yes	yes	d	690	$\sim 2 \cdot 10^9$
Independent Set	L / L	s	500	1	1	yes	yes	d	500	2
Network	L / L	s	1,211	10^6	5	yes	no	s	46,330	$\sim 2.5 \cdot 10^{19}$
Routing	L / L	s	-	-	< 10	yes	no	s	-	-
Topology	L / Q	s	2,176	2	$4 - 7$	yes	no	s	-	-

Zur Seite 15

Type (Obj/Constraints): F = Feasibility L = Linear, Q = Quadratic objective
It is to note that quadratic can be linearized, abs can be achieved by squaring, etc.
Small instances of sparse problems might be dense. By modelling in certain ways problems can also become dense.
Coefs is the maximum size of coefficients assuming everything scaled to integer.
Feas: “yes” means it is trivial to find a feasible solution.
Bin: Pure binary: “yes” means non-binary variables are not helpful.

Instance properties



Fast(est) Solvers

- **QuBowl: Quadratic Unconstraint Binary Optimization (QUBO)**
  Faster exact solution of sparse MaxCut and QUBO problems
  Rehfeldt, Koch, Shinano, 2023, arXiv:2202.02305 (to appear in MPC)
  http://plato.asu.edu/ftp/qubo.html 46% faster than the next best!

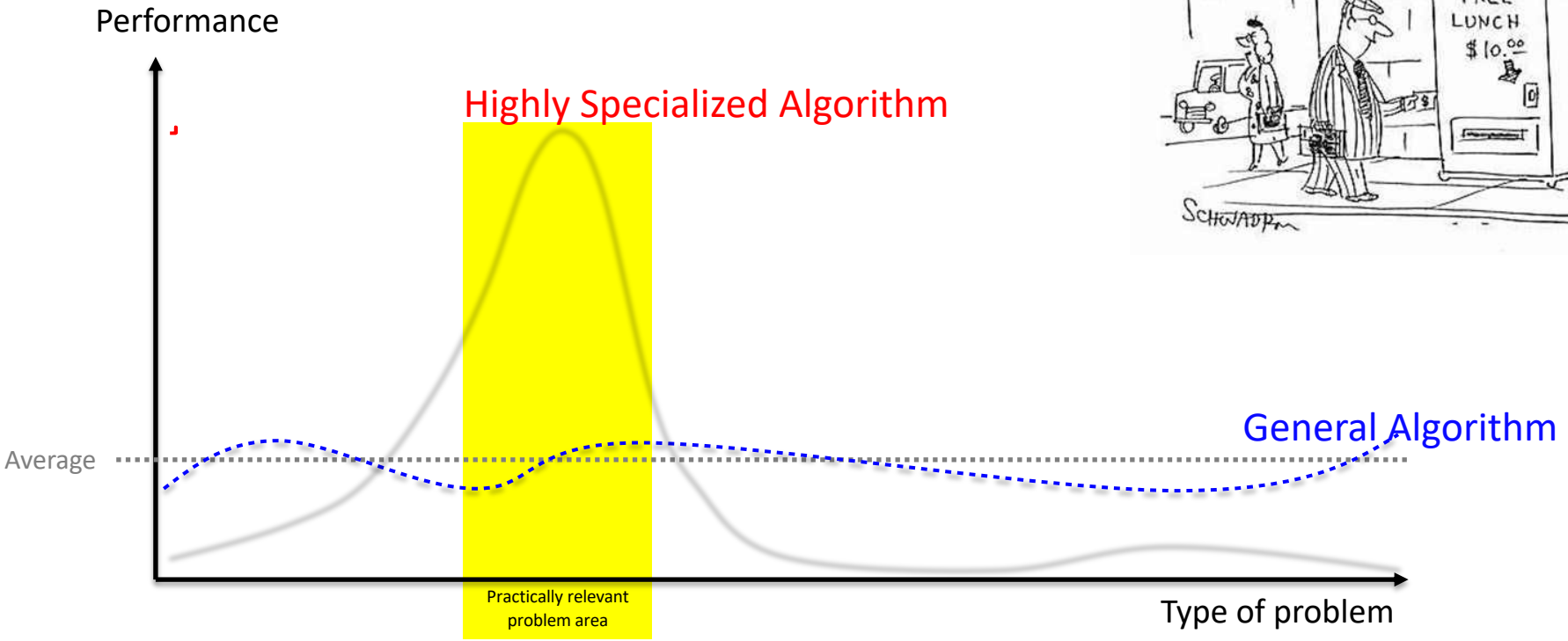
- **SCIP-Jack: Steiner Tree Problems in Graphs and variants (STPG)**
  Implications, conflicts, and reductions for Steiner trees
  Rehfeldt, Koch, 2023, MPC, doi: 10.1007/s10107-021-01757-5
  https://scipjack.zib.de Orders of magnitude faster then most others.

- **PIPS++: Block Structured Linear Programs with coupling constraints and Variables**
  A massively parallel interior-point solver for LPs with generalized arrowhead structure, and applications to energy system models
  Rehfeldt, Hobbie, Schönheit, Koch, Möst, Gleixner, 2022, EJOR, doi: 10.1016/j.ejor.2021.06.063
  https://github.com/NCKempke/PIPS-IPMpp >20 x faster than all commercial solvers

- **SCIP: LP/MILP/MINLP and more**
  The SCIP Optimization Suite 8.0
  Bestuzheva et. al., 2022, https://optimization-online.org/2021/12/8728
  https://www.scipopt.org Fastest Open-Source MILP/MINLP solver

TANSTAAFL or trading generality for speed

There is an uncountable infinite number of problems,
while there is only a countable infinite number of instances we practically want to solve.



https://en.wikipedia.org/wiki/No_free_lunch_in_search_and_optimization
https://towardsdatascience.com/a-blog-about-lunch-and-data-science-how-there-is-no-such-a-thing-as-free-lunch-e46fd57c7f27

SCIP-Jack: Why not using a general MIP solver?

Consider (small-scale) network design instance with:

|V| = 12,715, |E| = 20,632, |T| = 475

Using generic flow formulation with CPLEX 12.10/Gurobi 9.0: cannot be solved within 48 hours.

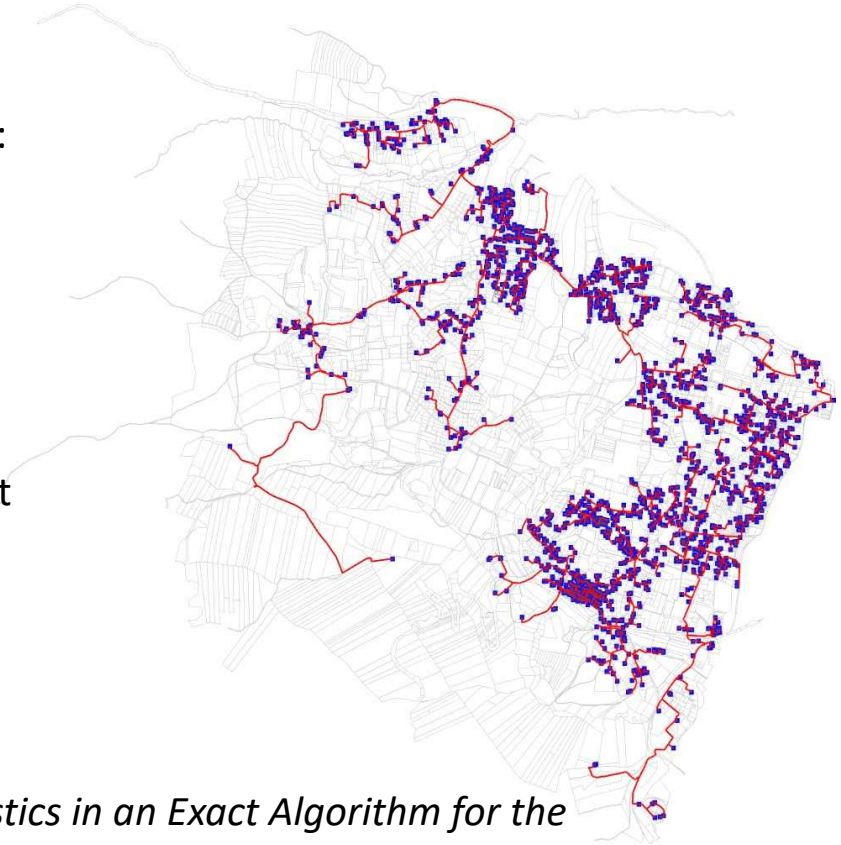
SCIP-Jack: Solves to optimality in 0.3 seconds

Largest (real-world) instance solved by SCIP-Jack so far

|E| = 32,388,930.

For larger problems CPLEX/Gurobi run out of memory almost immediately with 64 GB RAM.

Network telecommunication design for Austrian cities, see *New Real-world Instances for the Steiner Tree Problem in Graphs* (Leitner et al., 2014)



Combining NP-Hard Reduction Techniques and Strong Heuristics in an Exact Algorithm for the Maximum-Weight Connected Subgraph Problem, Rehfeldt, K., 2019, doi: 10.1137/17M1145963

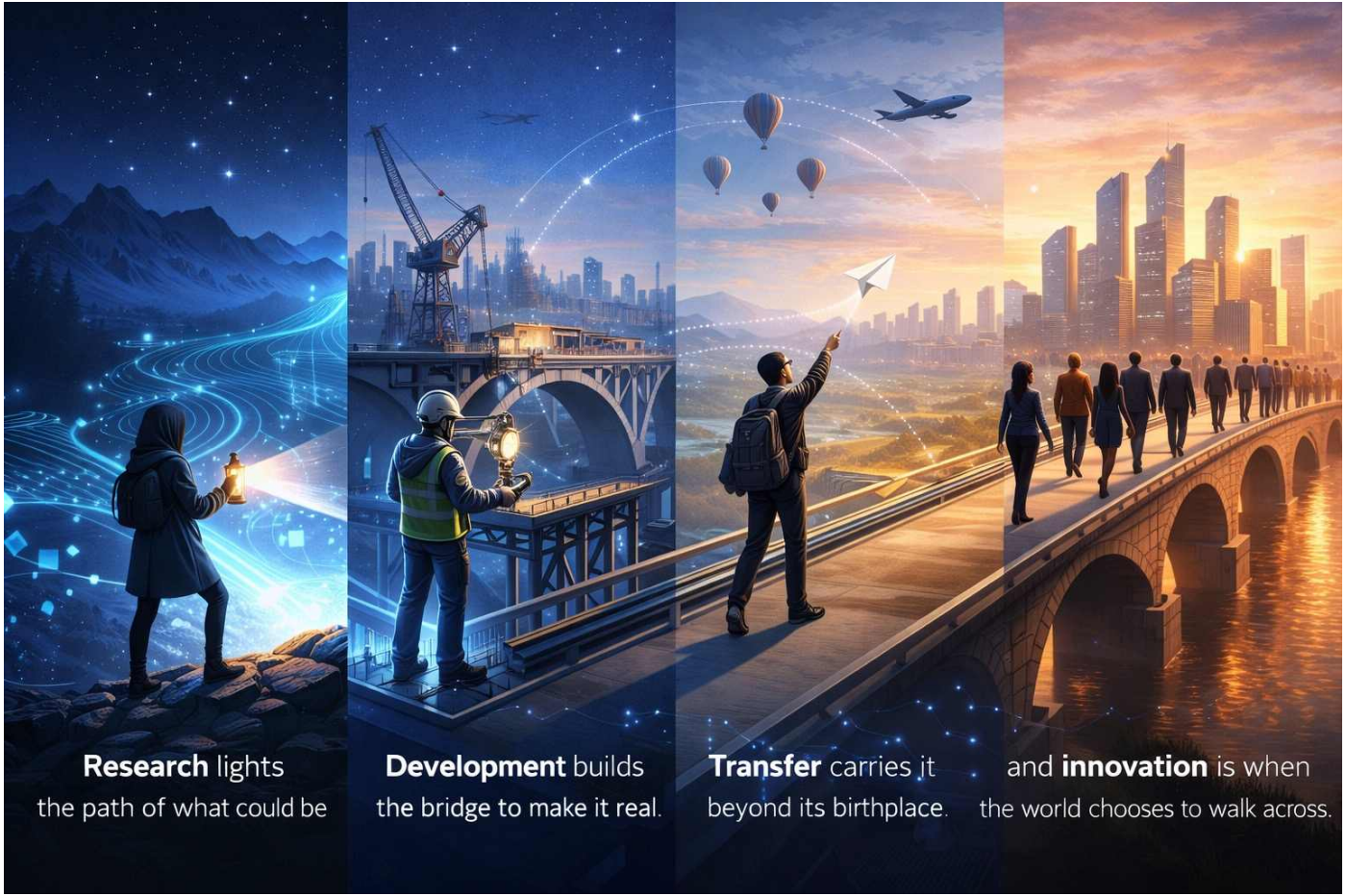
AUT VIAM INVENIAMUS AUT FACIAMUS

Research discovers what is possible,

development makes it work,

translation spreads it to new contexts,

and innovation is when it is actually adopted to create value.



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Vice President of the Bachelier Finance Society (2024)
Ecole Polytechnique

IntelligenceX 2026 Conference: The Global Quantum×AI Frontier

Since 2022, I have been organizing a series of **quantum computing events** focused on real-world applications and industry engagement.

- *Frontiers in FinTech and Quantum Computing* (2022),
- *Accelerating Industrial Innovation with Quantum Computing and Data Science* (2023),
- *Quantum Computing Workshop – AI, Optimization, and Forecasting Across Industries: Digital and Quantum Computing* (2024),
- *ETH–NUS AI×FinTech Workshop and Hackathon* (2025) and
- *ETH Zürich Fintech & Quantum Finance Workshop (2025 in Zurich)* *and* *The 9th ISM-ISCT-NII-ZIB-NUS-MODAL Workshop on Optimization and Machine Learning for Data Science and Future Computing (2025 in Tokyo)*

Each event has brought together **industry technology leaders and academic researchers**, combining tutorials, hackathons, and/or collaborative discussions to bridge research and practice.

We are planning a **3-day Quantum×AI Conference on 24-26 and 28 September 2026**. The event will feature:

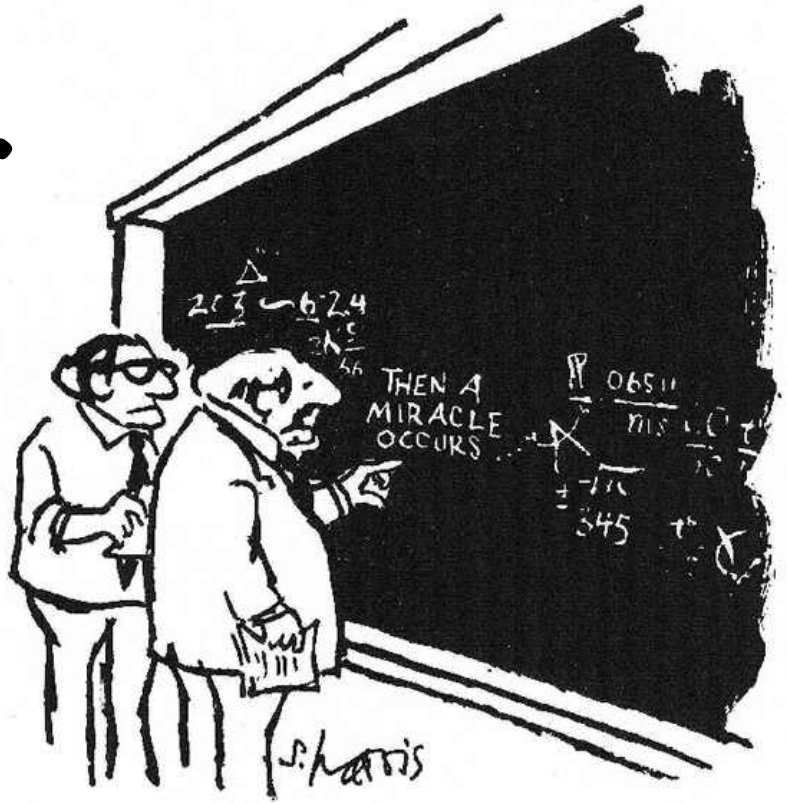
- **Keynotes** by leading researchers from academia and industry
- **Parallel sessions** on quantum computing and AI foundations and applications
- **Hands-on tutorials** on quantum fundamentals, optimization, forecasting, and hybrid QxAI workflows supported with user cases mainly in finance, transportation and healthcare. Public training sessions for practitioners and researchers on quantum computing and AI foundations and implementations.
- **Panel discussions and roundtables** fostering academic–industry collaboration
- **Ecosystem forum**: industry–academia showcases and collaboration opportunities.
- **An executive track (by invitation only)** designed for senior technology and innovation leaders, offering strategic briefings and curated roundtables for decision-makers

“This is only a foretaste of what is to come and only the shadow of what is going to be”

— Alan Turing, 1949

Research to encounter new knowledge

Questions?



"I think you should be more explicit here in step two."